Qualcomm

Cooperation between CPU and system level cache by using MPAM

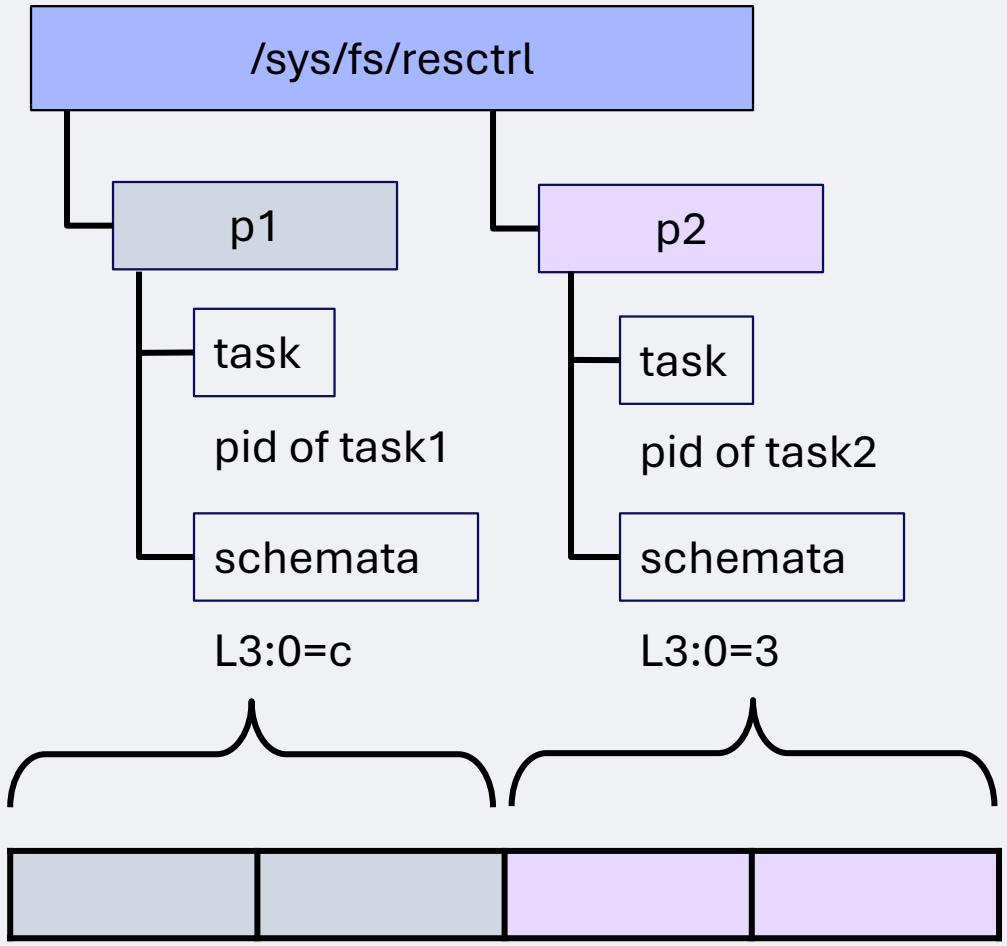
Yiwei Huang
huang.yiwei@oss.qualcomm.com
Qualcomm CSE Linux kernel team



Snapdragon and Qualcomm branded products are products of Qualcomm Technologies, Inc. and/or its subsidiaries.
Qualcomm patented technologies are licensed by Qualcomm Incorporated.

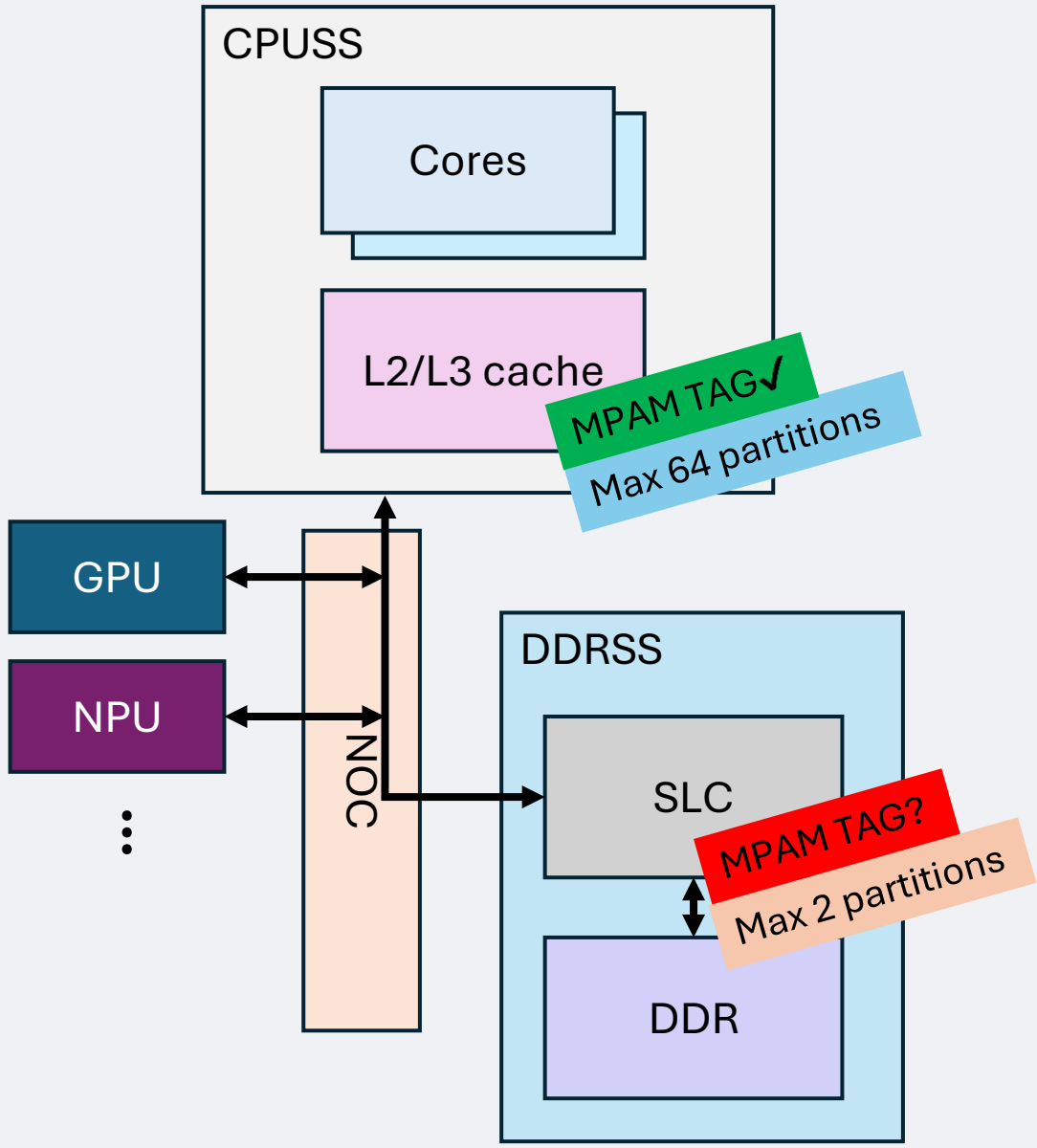
Background

- MPAM (Memory System Resource Partitioning and Monitoring)
  - Enables cache partitioning using bitmaps and maximum percentage limits
  - Provides memory bandwidth control for QoS
- Resctrl filesystem
  - A standardized user interface for resource allocation and monitoring
  - Originally introduced for Intel® RDT (Resource Director Technology)
  - ARM is adopting resctrl for MPAM, ensuring cross-architecture consistency



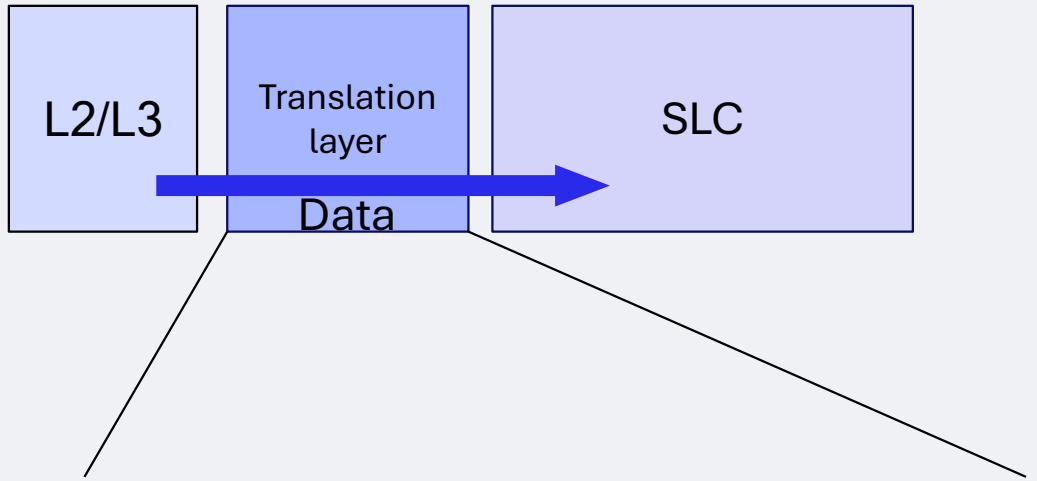
Problems

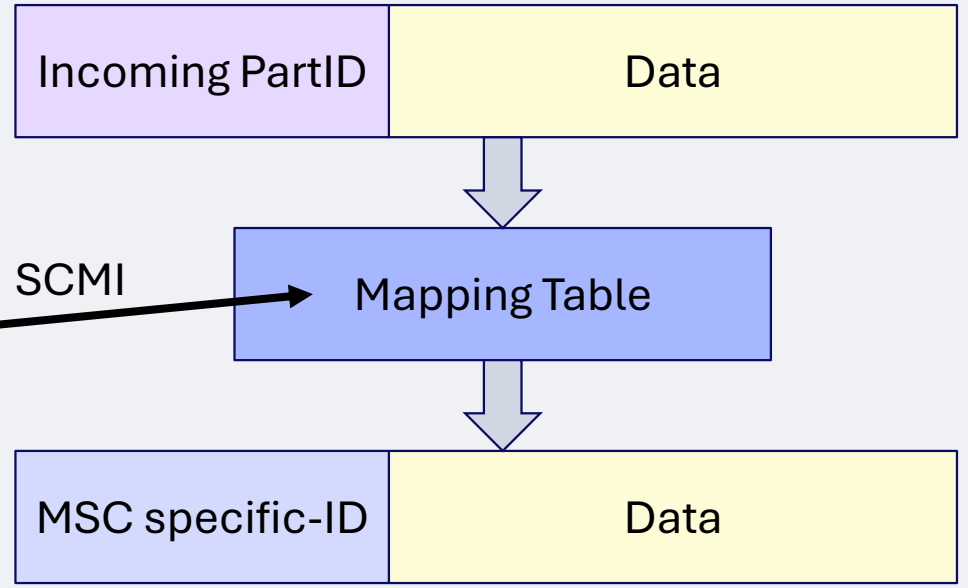
- Some MSCs are not fully compliant with the MPAM specification
- SLC (System Level Cache) is originally client-based and can simulate a limited number of partitions
- System-wide partition number is constrained by the lowest hardware-supported partition capability



Solution

- Introduce a translation layer before the MSC which does not natively support MPAM
- Convert incoming PartID into an MSC-specific ID for compatibility
- Add a new resource type called mapping in resctrl to allow users to configure the mapping table

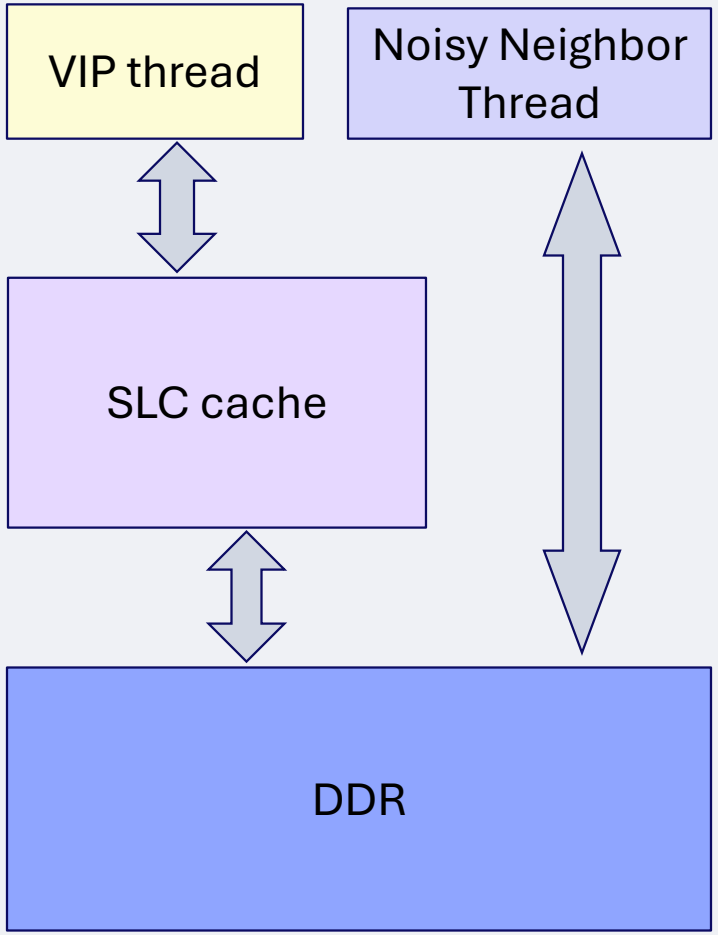




Incoming PartID	MSC-specific ID
1	1
2	1
3	2
…	…

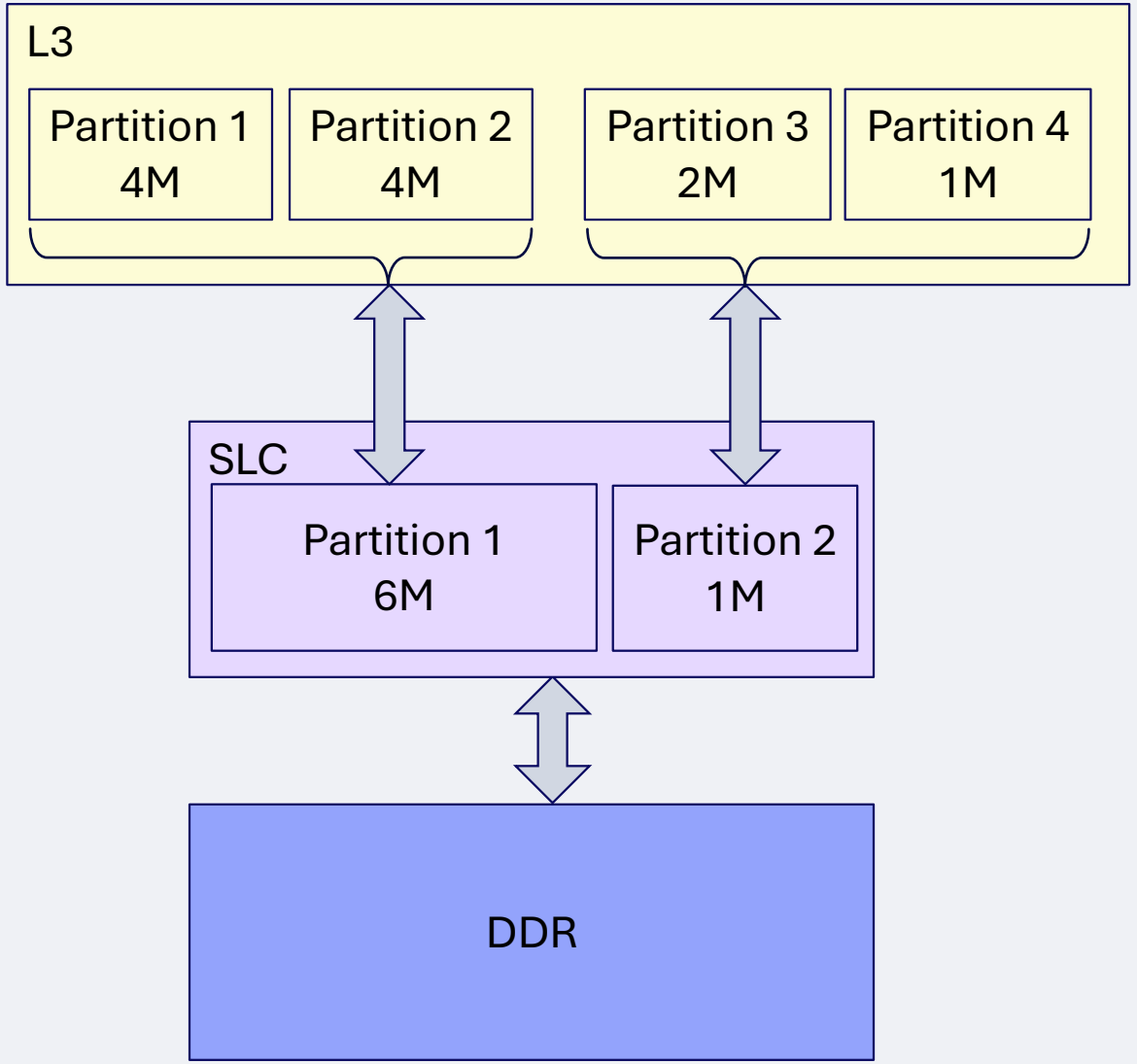
Use Cases

- A simple case is that a noisy neighbor thread can completely bypass the SLC
- Background download and software upgrade



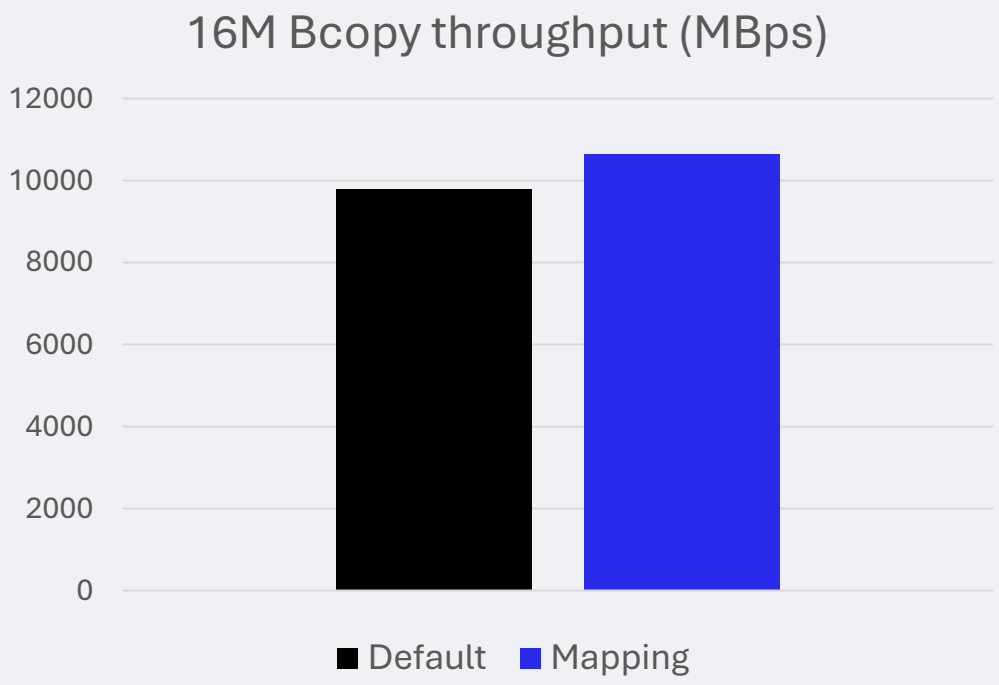
Use Cases

- A more complex case involves using multiple partitions and mapping them to different partitions on the SLC
- Parallel workload on Foldable Screen Devices
- High-Performance Gaming Scenarios



Unit Test Example

- Two bw_mem tasks run in parallel — one as a benchmark, the other as an interference workload
- Default case: Both tasks share SLC, simulating interference
- Mapping case: Interference task excluded from SLC usage
- Benchmark task performance improves by ~8%



Thank you

Nothing in these materials is an offer to sell any of the components or devices referenced herein.

© Qualcomm Technologies, Inc. and/or its affiliated companies. All Rights Reserved.

Qualcomm and Snapdragon are trademarks or registered trademarks of Qualcomm Incorporated.
Other products and brand names may be trademarks or registered trademarks of their respective owners.

References in this presentation to “Qualcomm” may mean Qualcomm Incorporated, Qualcomm Technologies, Inc., and/or other subsidiaries or business units within the Qualcomm corporate structure, as applicable. Qualcomm Incorporated includes our licensing business, QTL, and the vast majority of our patent portfolio. Qualcomm Technologies, Inc., a subsidiary of Qualcomm Incorporated, operates, along with its subsidiaries, substantially all of our engineering, research and development functions, and substantially all of our products and services businesses, including our QCT semiconductor business.

Snapdragon and Qualcomm branded products are products of Qualcomm Technologies, Inc. and/or its subsidiaries. Qualcomm patented technologies are licensed by Qualcomm Incorporated.

Follow us on:
For more information, visit us at qualcomm.com & qualcomm.com/blog

We are Hiring Linux Kernel Engineers

San Diego

tsoni@quicinc.com

Linux maintainers(remote)

ndechesn@quicinc.com

Hyderabad

China

QIPL